TEXT OR AUDIO TO SIGN LANGUAGE CONVERSION

A COMMUNITY FOCUS PROJECT REPORT

216ECE4221

Submitted by

B.Akhileswara Reddy (9922005345)
C.Ganesh (9922005386)
V.Sai Kiran(9922005413)
B.Jagan Mohan Reddy(9922005411)

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SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

KRISHNANKOIL 626 126

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Major Design Experience Information

Student Details	: B.Akhileswara Reddy C.Ganesh B.Jagan Mohan Reddy V.Sai kiran
Project Supervisor	: Dr.K.S.Dhanalakshmi
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Anand Nagar, Krishnankoil, Srivilliputtur (Via), Virudhunagar (Dt) - 626126, Tamil Nadu | info@kalasalingam.ac.in | www.kalasalingam.ac.in



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We hereby declare that this project **"Text Or Audio To Sign Language Conversion"** is our genuine work and no part of it has been reproduced from any other works.

REGISTER. No:	STUDENT NAME	STUDENT SIGNATURE
9922005345	B.Akhileswara Reddy	
9922005386	C.Ganesh	
9922005413	V.Sai Kiran	
9922005411	B.Jagan Mohan Reddy	

Date:

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BONAFIDE CERTIFICATE

Certified that this project report on "**Text Or Audio To Sign Language Conversion**" is the work of **"B.AkhileswaraReddy(9922005345),C.Ganesh(9922005386),V.Saikiran(9922005386),B.JaganMohn Reddy(9922005411)"** who carried out the project work under my supervision.

SIGNATURE
Dr.K.S.Dhanalakshmi M.Tech,Ph.D
Associate Professor,
Department of ECE
Kalasalingam Academy of
Research and Education
Krishnankoil-626126.

SIGNATURE
Dr.K.Jeya Prakash M.E,Ph.D
Associate Professor,
Department of ECE
Kalasalingam Academy of
Research and Education
Krishnankoil-626126

SIGNATURE
Dr.A. Muthukumar, ME., Ph.D.,
Head of the Department
Associate Professor
Department of ECE
Kalasalingam Academy of
Research and Education
Krishnankoil-626126.

Project Final Review Viva-voce held on ……

Internal Examiner **External Examiner**

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ACADEMY OF RESEARCH AND EDUCATION
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Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade
Anand Nagar, Krishnankoil - 626126. Srivilliputtur (Via), Virudhunagar (Dt), Tamil Nadu | info@kalasalingam.ac.in | www.kalasalingam.ac.in



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE FROM THE COMMUNITY MEMBER(S) (If ANY)

Certified that this project "Text Or Audio To Sign Language Conversion" is the work of "**B.Akhileswara Reddy(9922005345),C.Ganesh(9922005386),V.Sai kiran(9922005413)B.Jagan Mohan Reddy(9922005411)""**" who carried out the survey of our community and provided the necessary solution to our problem.

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ABSTRACT

This paper introduces a new system that converts text or audio inputs into sign language animations, thus enabling an accessible means of communication for people who are deaf or hard-of-hearing. The system tokenizes words in text or transcribed audio, maps each token to its corresponding sign language gesture, and renders these gestures as animated sequences. Unlike the traditional static dictionaries for sign language, this system is based on NLP and animation frameworks. Such an application is likely to help in smooth communication among the different domains: educational settings, social spheres, and digital worlds. In this context, we outline the methodology, technical challenges, and possible applications of the system. It gives rise to a critical communication accessibility need for a population depending on sign language. Through text or spoken language transformation into sign language animations, the system opens up new avenues of interaction across divergent contexts. This research contributes to literature through the provision of an approach scalable and automated which may be adapted across the various languages and dialects of sign language, therefore pushing the boundaries further on the side of inclusiveness in technology

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CHAPTER 1

INTRODUCTION

Sign language is a crucial means of communication for deaf and hard-of-hearing populations with its grammar, syntax, and cultural nuances within a structured, visual language. However, there remains a communication gap between the users of sign language and the users of spoken or written language. This would pose a problem in the normal course of life, in educational opportunities, and at work.Then the possibility exists in attempting to bridge this gap between speech and sign language by adding natural language. processing, married with advances in animation made possible through automation as automated systems convert text and audio to animated signs. Then there will be visual ways by which more will enjoy language, both in terms of spoken and written. Thus, it would become both more accessible, communicable, and thereby enjoyable. It enables an automatic conversion system to interpret at runtime either text or audio inputs, tokenize them into individual words or phrases, and then present them as animated sequences. This approach can therefore better exhibit and support natural and flexible interaction flows like live presentations. This research addresses multiple technical difficulties, such as the exact input of context and idioms, managing in real time the processing necessities as well as representing the complexity of sign language through all its subtle animative traits. By elaborating from these components with utmost analysis, the study brings itself to contribution in that growing field known as access technology, between sign languages users and broader populations as well.

CHAPTER 2

LITERATURE SURVEY

TITLE OF THE PAPER	ALGORITHM USED	YEAR OF PUBLICATION
Sign Language Translation	Rule-based and machine learning approaches	2020
Sign Language Translation: A Comparative Study of Deep Learning Approaches	Deep learning models	2020
Real-Time Speech Recognition for Sign Language Generation	Applied speech recognition with recurrent neural networks (RNN)	2019
Sign Language Animation Synthesis for Text-to-Sign Language Translation	Used NLP for language parsing and animation software to visualize signs	2020
Text and Audio-Based Communication System for Deaf Communities	Combined NLP for text, speech recognition for audio, and GANs for generating sign gestures	2022

CHAPTER 3

METHODOLOGY

The methodology for developing a text and audio-to-sign language conversion system involves several key steps: processing the input, mapping to appropriate sign language gestures, generating animations, and testing for accuracy and usability. Each phase incorporates specialized technologies and techniques to ensure accurate, real-time, and context-sensitive sign language translations.

Input Processing and Tokenization:

Text Input: The system first takes text input and applies tokenization, which breaks down sentences into individual words or phrases. This process is guided by linguistic rules tailored for sign language, considering that sign language often omits certain articles and auxiliary words. By focusing on essential terms and contextual nuances, the tokenization algorithm ensures a more natural translation into sign language structure.

Audio Input: For audio, the system incorporates an automatic speech recognition (ASR) model that transcribes spoken language into text. This transcription is then passed through the tokenization process. ASR accuracy and speed are optimized to allow for real-time processing, which is essential for seamless interaction in conversational settings.

Contextual Analysis: Both text and audio inputs undergo contextual analysis to understand idiomatic expressions, grammatical structures, and other nuances that may affect the meaning in sign language. Natural language processing (NLP) techniques, such as named entity recognition and part-of-speech tagging, help identify key elements in sentences that require special handling during translation.

Gesture Database: Once tokenized, each word or phrase is matched with a corresponding sign language gesture stored in a database. This database comprises a range of sign language gestures, accounting for synonyms, variations, and context-specific expressions to ensure accuracy in diverse scenarios.

Synonym and Contextual Matching: The system uses a synonym matching algorithm to select the closest corresponding gesture if an exact match for a word is unavailable. This approach helps manage cases where direct translations may not exist, particularly for specialized or technical terms.

Grammar Adjustments: Sign language grammar differs significantly from spoken language grammar, often following a subject-object-verb structure. The system makes adjustments based on these rules, rearranging the sequence of gestures if needed to reflect proper sign language

grammar.

CHAPTER 4

SOFTWARE

DESCRIPTION:

A proposed system for "Text or Audio to Sign Language Conversion," is a web application developed for the conversion of spoken as well as written languages into equivalent signs in sign language. The application is implemented with a good framework that makes Django in the process of the back-end, Tensor flow for deep learning models, and the library of 3D animation for generating gestures.

Front-end Interface

User Input Interface: The user can input their text or speak directly into the system. The interface provides a text-box and audio recording functionality that can support both text and speech input options.

Real-Time Display: On completion of input processing, the output is presented as animated sign language gestures. The 3D avatar or a video-based sign language sequence renders gestures in real-time. This way, the user gets to see a very visual and engaging experience.

Backend Framework

API Integration: Django's REST framework communicates smoothly between the front end, NLP and ASR models, and gesture generation modules. APIs manage requests and handle data processing, ensuring consistent interaction across system components.

Database

The system uses a database such as PostgreSQL or MySQL to store session data, input histories, and logs, which can be monitored in order to determine system problems.

Text Preprocessing Module

- Tokenization: The text input is segmented into individual words or phrases, making it easier for the model to process language constructs.
- Lemmatization and Stemming: This step reduces words to their root forms, enhancing the model's ability to generalize and identify word meanings consistently.
- Stop Word Removal: Common but non-essential words are filtered out, reducing noise and improving processing efficiency without losing the intended meaning.

Speech Recognition Module

- Automatic Speech Recognition (ASR): For audio input, ASR technology (such as Deep Speech) converts spoken words into text. The ASR component uses deep neural networks to recognize spoken language, translating it into text that is fed into the NLP pipeline.

Noise Reduction and Speech Segmentation: Preprocessing techniques are used to remove background noise and segment speech, allowing the system to operate effectively even in less-than-ideal audio conditions

Translation and Contextualization Module

- Contextual NLP Model: This model processes text inputs to understand context, sentiment, and syntax, mapping words and phrases accurately to sign language structures. This is crucial as sign languages have different grammatical structures and may require rearrangement of words or phrases.
- Grammar Conversion for Sign Language: NLP rules and techniques are applied to restructure sentences according to the grammatical norms of sign language, ensuring coherent and natural translation. This involves rearranging sentence components to match the syntax of the target sign language.

Sign Language Gesture Mapping

- NLP-to-Gesture Mapping: Each word or phrase processed by the NLP model is mapped to a corresponding sign gesture. The mapping is handled by an internal dictionary of sign language gestures, ensuring that each translated word or phrase is accurately represented in sign language.

CHAPTER 5

IMPLEMENTATION

Program coding :

```
from django.http import HttpResponse
from django.shortcuts import render, redirect
from django.contrib.auth.forms import UserCreationForm, AuthenticationForm
from django.contrib.auth import login,logout
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
import nltk
from django.contrib.staticfiles import finders
from django.contrib.auth.decorators import login_required

def home_view(request):
    return render(request,'home.html')

def about_view(request):
    return render(request,'about.html')

def contact_view(request):
    return render(request,'contact.html')

@login_required(login_url="login")
def animation_view(request):
    if request.method == 'POST':
        text = request.POST.get('sen')
        #tokenizing the sentence
        text.lower()
        #tokenizing the sentence
        words = word_tokenize(text)

        tagged = nltk.pos_tag(words)
        tense = {}
        tense["future"] = len([word for word in tagged if word[1] == "MD"])
```

```
        tense["present"] = len([word for word in tagged if word[1] in ["VBP", "VBZ","VBG"]])
        tense["past"] = len([word for word in tagged if word[1] in ["VBD", "VBN"]])
        tense["present_continuous"] = len([word for word in tagged if word[1] in ["VBG"]])


        #stopwords that will be removed
        stop_words = set(["mightn't", 're', 'wasn', 'wouldn', 'be', 'has', 'that', 'does', 'shouldn', 'do', "you've",'off',
'for', "didn't", 'm', 'ain', 'haven', "weren't", 'are', "she's", "wasn't", 'its', "haven't", "wouldn't", 'don', 'weren', 's',
"you'd", "don't", 'doesn', "hadn't", 'is', 'was', "that'll", "should've", 'a', 'then', 'the', 'mustn', 'i', 'nor', 'as', "it's",
"needn't", 'd', 'am', 'have',  'hasn', 'o', "aren't", "you'll", "couldn't", "you're", "mustn't", 'didn', "doesn't", 'll',
'an', 'hadn', 'whom', 'y', "hasn't", 'itself', 'couldn', 'needn', "shan't", 'isn', 'been', 'such', 'shan', "shouldn't", 'aren',
'being', 'were', 'did', 'ma', 't', 'having', 'mightn', 've', "isn't", "won't"])
lr = WordNetLemmatizer()
        filtered_text = []
        for w,p in zip(words,tagged):
            if w not in stop_words:
                if p[1]=='VBG' or p[1]=='VBD' or p[1]=='VBZ' or p[1]=='VBN' or p[1]=='NN':
                    filtered_text.append(lr.lemmatize(w,pos='v'))
                elif p[1]=='JJ' or p[1]=='JJR' or p[1]=='JJS'or p[1]=='RBR' or p[1]=='RBS':
                    filtered_text.append(lr.lemmatize(w,pos='a'))

                else:
                    filtered_text.append(lr.lemmatize(w))

        #adding the specific word to specify tense
        words = filtered_text
        temp=[]
        for w in words:
            if w=='I':
                temp.append('Me')
            else:
                temp.append(w)
        words = temp
        probable_tense = max(tense,key=tense.get)

        if probable_tense == "past" and tense["past"]>=1:
            temp = ["Before"]
            temp = temp + words
            words = temp
        elif probable_tense == "future" and tense["future"]>=1:
            if "Will" not in words:
                    temp = ["Will"]
                    temp = temp + words
                    words = temp
            else:
                pass
        elif probable_tense == "present":
            if tense["present_continuous"]>=1:
                temp = ["Now"]
                temp = temp + words
                words = temp

        filtered_text = []
        for w in words:
            path = w + ".mp4"
            f = finders.find(path)
```

```
            #splitting the word if its animation is not present in database
            if not f:
                for c in w:
                    filtered_text.append(c)
            #otherwise animation of word
            else:
                filtered_text.append(w)
        words = filtered_text;

        return render(request,'animation.html',{'words':words,'text':text})
    else:
        return render(request,'animation.html')

def signup_view(request):
    if request.method == 'POST':
        form = UserCreationForm(request.POST)
        if form.is_valid():
            user = form.save()
            login(request,user)
            # log the user in
            return redirect('animation')
    else:
        form = UserCreationForm()
    return render(request,'signup.html',{'form':form})


def login_view(request):
    if request.method == 'POST':
        form = AuthenticationForm(data=request.POST)
        if form.is_valid():
            #log in user
            user = form.get_user()
            login(request,user)
            if 'next' in request.POST:
                return redirect(request.POST.get('next'))
            else:
                return redirect('animation')
    else:
        form = AuthenticationForm()
    return render(request,'login.html',{'form':form})

def logout_view(request):
    logout(request)
    return redirect("home")
```

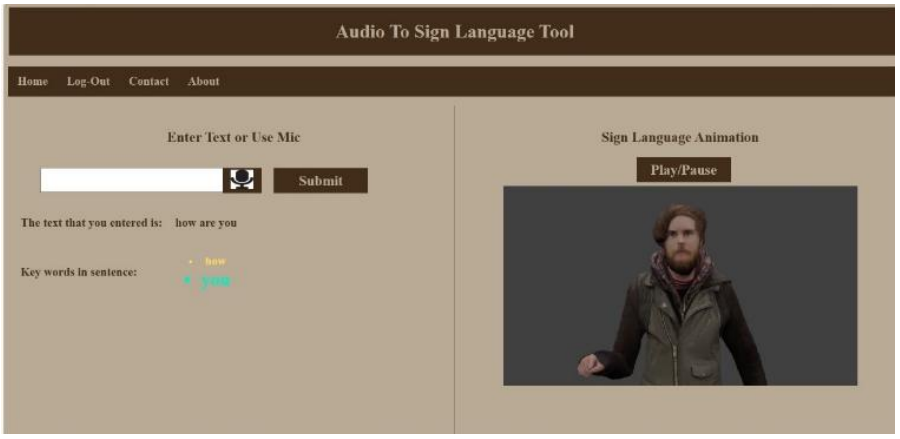
Out Put:



Login Page



Interface of input of Text/Audio



Output of text/audio in the form of animation

CHAPTER 6

CONCLUSION:

The conversion of text or audio to sign language is a critical technological advancement aimed at bridging communication gaps between the hearing and deaf communities. By using sophisticated algorithms and machine learning models, these systems can translate spoken or written language into sign language in real-time, facilitating more inclusive interactions and improving accessibility. While significant progress has been made in developing such systems, challenges remain in ensuring accuracy, context understanding, and regional variations of sign languages. Future research and development will likely focus on enhancing the naturalness of sign language translations, refining real-time performance, and promoting widespread adoption of these tools to create a more inclusive world for individuals who are deaf or hard of hearing.

CHAPTER 7

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